

AI ASSISTED CODING

Lab Experiment: Documentation Generation -Automatic documentation and code comments

Problem 1 – find_max(numbers)

(a) Docstring Style

```
def find_max(numbers):  
    """  
    Returns the largest number from a list.  
  
    Parameters:  
        numbers (list): A list of numbers.  
  
    Returns:  
        int/float: The largest number in the list.  
    """  
    return max(numbers)
```

(b) Inline Comments

```
def find_max(numbers):  
    # Find and return the maximum value  
    return max(numbers)
```

(c) Google-Style Documentation

```
def find_max(numbers):  
    """Returns the maximum value from a list.  
  
    Args:  
        numbers (list): List of numeric values.  
  
    Returns:  
        int | float: Maximum value in the list.  
    """  
    return max(numbers)
```

Problem 2 – Login Function

(a) Docstring

```
def login(user, password, credentials):
    """
    Checks whether a user's password matches the stored password.

    Parameters:
        user (str): Username.
        password (str): Password entered.
        credentials (dict): Dictionary of usernames and passwords.

    Returns:
        bool: True if login succeeds, otherwise False.
    """
    return credentials.get(user) == password
```

(b) Inline Comments

```
def login(user, password, credentials):
    # Compare entered password with stored password
    return credentials.get(user) == password
```

(c) Google Style

```
def login(user, password, credentials):
    """Validates user login credentials.

    Args:
        user (str): Username.
        password (str): Entered password.
        credentials (dict): Username-password dictionary.

    Returns:
        bool: Login status.
    """
    return credentials.get(user) == password
```

Problem 3 – Calculator Module (calculator.py)

Create a file named calculator.py

```
"""
```

Calculator Module

Provides basic arithmetic operations.

```
"""
```

```
def add(a, b):  
    """Returns the sum of two numbers."""  
    return a + b
```

```
def subtract(a, b):  
    """Returns the difference of two numbers."""  
    return a - b
```

```
def multiply(a, b):  
    """Returns the product of two numbers."""  
    return a * b
```

```
def divide(a, b):  
    """  
    Returns the quotient of two numbers.  
  
    Raises:  
        ZeroDivisionError: If b is zero.  
    """  
    return a / b
```

Display documentation in terminal
python -m pydoc calculator

Generate HTML documentation
python -m pydoc -w calculator

This creates:
calculator.html
Open it using your browser.

Problem 4 – Conversion Utilities (conversion.py)
Create conversion.py

```
"""
```

Conversion Utilities Module

Provides number conversion functions.

"""

```
def decimal_to_binary(n):
```

```
    """Converts a decimal number to binary."""
```

```
    return bin(n)[2:]
```

```
def binary_to_decimal(b):
```

```
    """Converts a binary string to decimal."""
```

```
    return int(b, 2)
```

```
def decimal_to_hexadecimal(n):
```

```
    """Converts a decimal number to hexadecimal."""
```

```
    return hex(n)[2:]
```

Generate documentation

Terminal:

```
python -m pydoc conversion
```

Generate HTML:

```
python -m pydoc -w conversion
```

Problem 5 – Course Management Module (course.py)

Create course.py

"""

Course Management Module

Stores and manages course information.

"""

```
courses = {}
```

```
def add_course(course_id, name, credits):
```

```
    """
```

```
        Adds a new course.
```

```
    Args:
```

```
        course_id (str): Course ID.
```

```
        name (str): Course name.
```

```

        credits (int): Credit value.
        """
        courses[course_id] = {
            "name": name,
            "credits": credits
        }

def remove_course(course_id):
    """
    Removes a course by ID.

    Args:
        course_id (str): Course ID.
    """
    courses.pop(course_id, None)

def get_course(course_id):
    """
    Returns course details.

    Args:
        course_id (str): Course ID.

    Returns:
        dict: Course information.
    """
    return courses.get(course_id)

```

Generate Documentation

View in terminal:

```
python -m pydoc course
```

Export HTML:

```
python -m pydoc -w course
```